



Natural Computing
Research & Applications
Group



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Investigating Hyperparameter Optimization and Transferability for ES-HyperNEAT: A TPE Approach

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ACM Workshop on

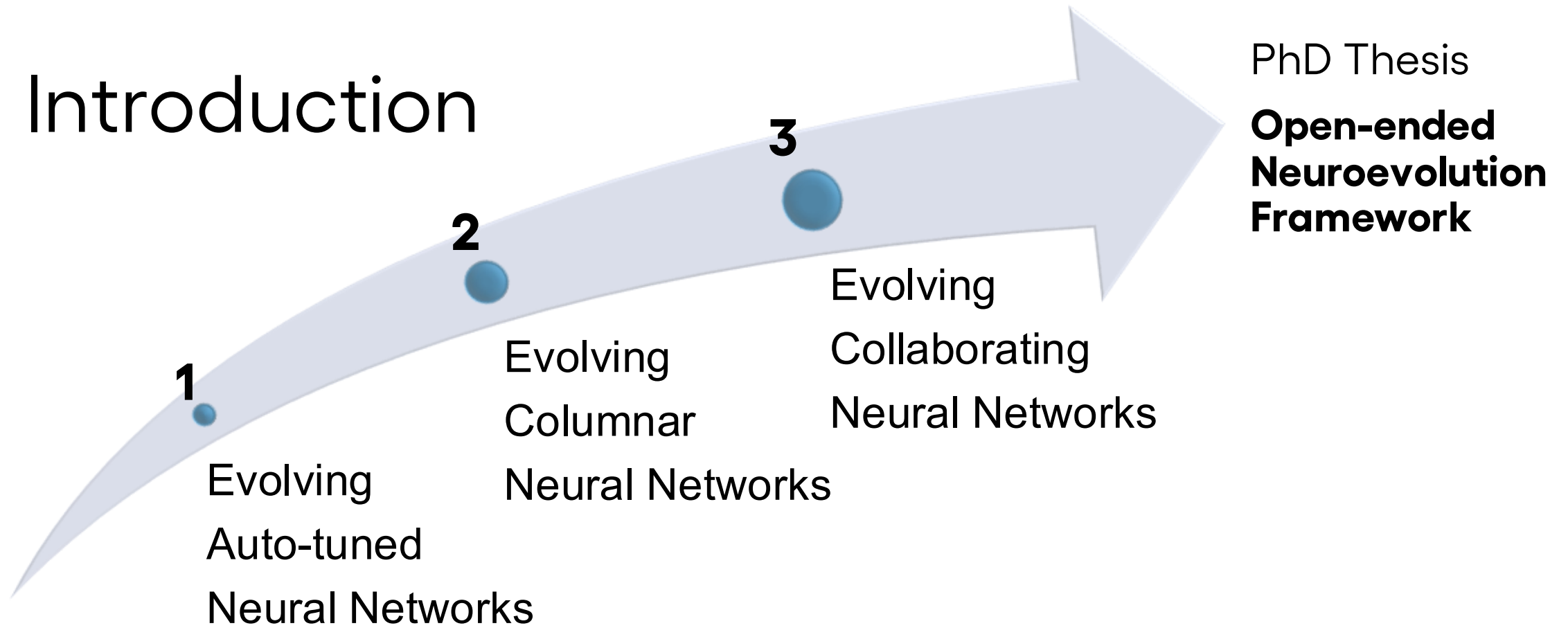
NeuroEvolution@Work

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Introduction

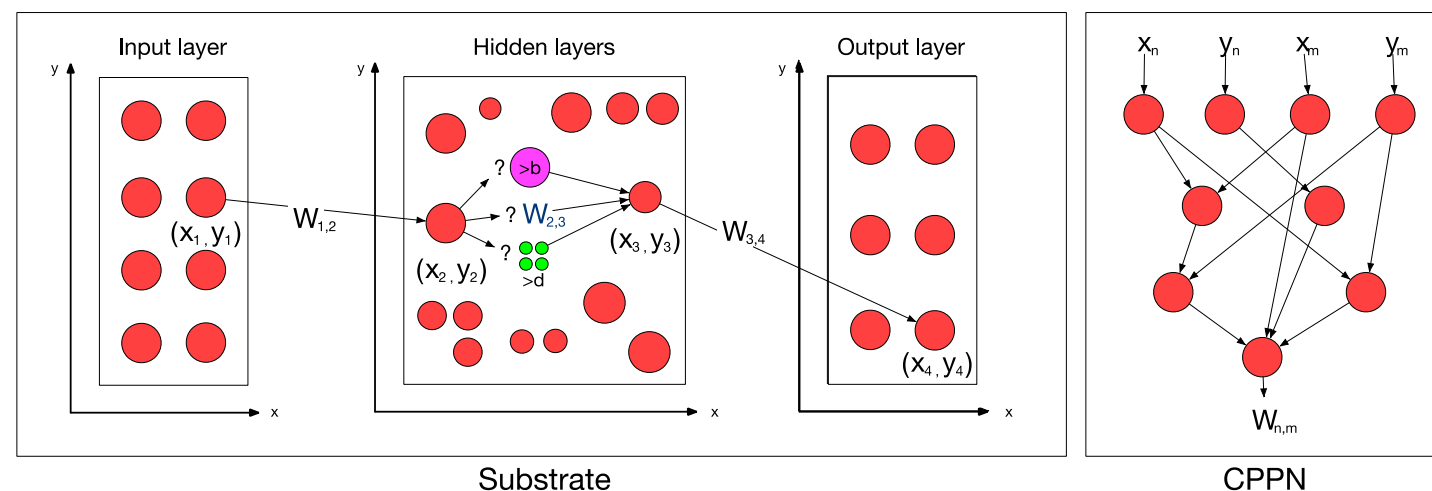


This study presents ongoing work on part 1:

- **ES-HyperNEAT** Hyperparameters Optimization using TPE on MNIST
- Transferring Hyperparameters found by TPE on MNIST to other tasks

Background

ES-HyperNEAT allows the substrate network topology to evolve



ES-HyperNEAT has about 70 hyperparameters for the CPPN and the Substrate

- Hyperparameters do significantly impact performance [1]

TPE is a Bayesian optimization algorithm for Hyperparameter Optimization [2]

- Modeling two probability distributions to estimate well-performing hyperparameters

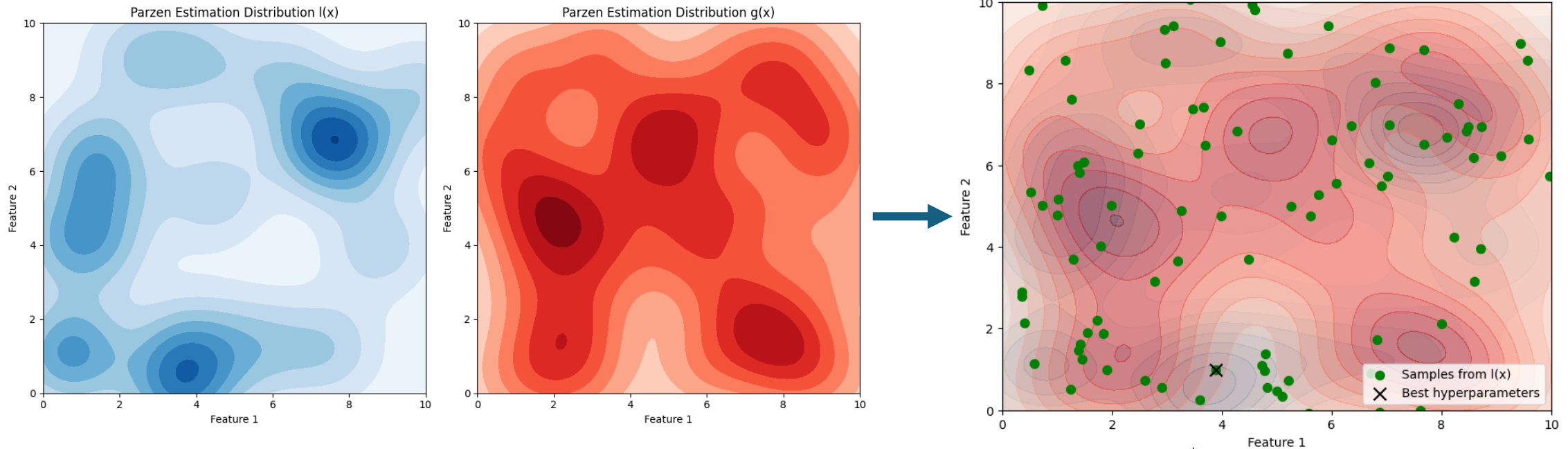
ES-HyperNEAT's default XOR Config

Hyperparameter	Value	Hyperparameter	Value
Substrate Parameters		NEAT Parameters	
initial_depth	0	fitness_criterion	max
max_depth	2	fitness_threshold	0.975
variance_threshold	0.03	pop_size	150
division_threshold	0.3	reset_on_extinction	False
max_weight	5.0	DefaultGenome Parameters	
iteration_level	1	activation_default	tanh
activation	sigmoid	activation_mutate_rate	0.5
		activation_options	gauss
			sin
			tanh
		aggregation_default	sum
		aggregation_mutate_rate	0.0
		aggregation_options	sum
		bias_init_mean	0.0
		bias_init_stdev	1.0
		bias_max_value	30.0
		bias_min_value	-30.0
		bias_mutate_power	0.5
		bias_mutate_rate	0.7
		bias_replace_rate	0.1
		:	:
		Network Parameters	
		num_inputs	5
		num_hidden_rate	3
		num_outputs	1
		:	:

Table 1: Example of substrate hyperparameters

Table 2: Example of CPPN hyperparameters

Tree-structured Parzen estimator (TPE)



TPE iteratively maintains two density estimators:

- $l(x)$ for the distribution with low objective values
- $g(x)$ for the distribution with high objective values

TPE efficiently explores the search space by selecting hyperparameters that maximize the expected improvement (EI) from then density estimators.

$$EI(\mathbf{x}) = \int_{-\infty}^{y^*} (y^* - y) \frac{l(\mathbf{x})}{g(\mathbf{x})} p(y|\mathbf{x}) dy$$

where y^* is the current best observed objective value, and $p(y | x)$ is the probability of observing an objective value y given the hyperparameters x .

Part 1 – Experimental Setup:

Hyperparameter Optimization on MNIST

- 16 Hyperparameters [3]
 - Search Space of over 3 billion combinations

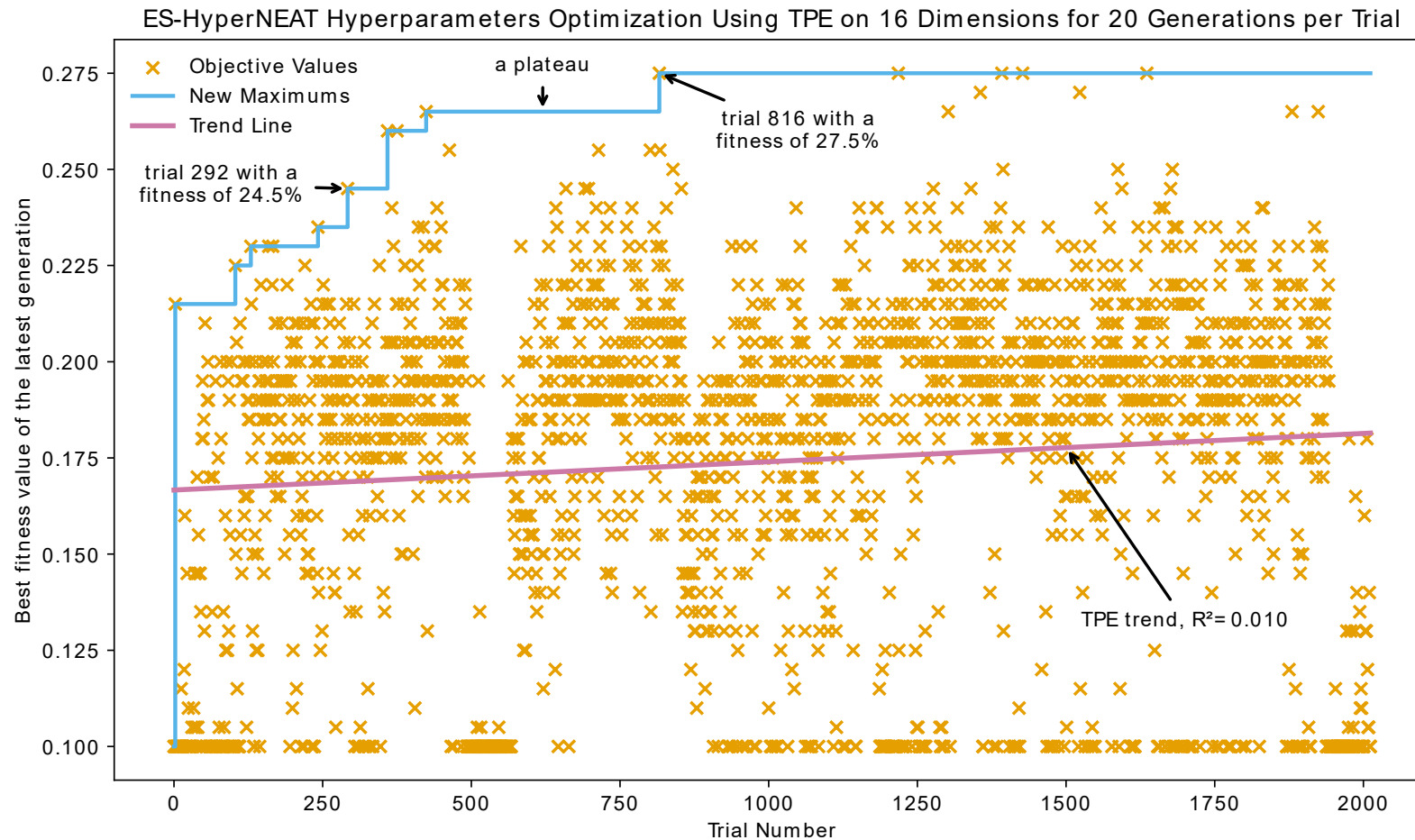
Hyperparameter	Range of Options
NEAT Parameters	
num_hidden	0, 1, 2, 3, 4, 5
pop_size	10, 20, 40, 60, 80, 100
conn_add_prob	0.3, 0.5, 0.7
conn_delete_prob	0.3, 0.5, 0.7
node_add_prob	0.2, 0.4, 0.6, 0.8
node_delete_prob	0.2, 0.4, 0.6, 0.8
initial_connection	full_nodirect, full_direct, unconnected, fs_neat_hidden, fs_neat_nohidden
connection_fraction	0.2, 0.4, 0.6, 0.8
activation_default	sigmoid, gauss, clamped, relu, tanh, softplus
Substrate Parameters	
initial_depth	1, 2
max_depth	3, 4, 5
variance_threshold	0.01, 0.02, 0.03, 0.04
division_threshold	0.3, 0.5, 0.7
max_weight	3.0, 5.0, 7.0
iteration_level	0, 1, 2, 3
activation	sigmoid, gauss, clamped, relu, tanh, softplus

Table 3: Hyperparameter search space for the Neuroevolution of Augmenting Topologies (NEAT) algorithm and substrate parameters.

- **TPE** vs. **Random Search**
- Default CPPN parameters:
 - XOR Config file
- 20 generations per trial
- 200 random images per gen
 - 20 x 10 classes
- Constraints
 - TPE: 78 days of computation
 - Random Search: 292 trials
- Latest Fitness as Objective Value
- Substrate coordinates centered on the input and output vectors

Part 1 – Experimental Results

TPE Optimization Results on MNIST



Comparison	t-statistic	p-value	Cohen's d
Hyperparameters Optimization for the MNIST Classification			
Random Search (292 trials)			
vs. Random Search (30 trials)	-0.10	0.922	-0.02
TPE Search (292 trials)			
vs. TPE Search (30 trials)	9.76	< 0.001	1.21
TPE Search (2013 trials)			
vs. TPE Search (30 trials)	14.29	< 0.001	1.55
TPE Search (2013 trials)			
vs. TPE Search (292 trials)	5.80	< 0.001	0.36
TPE Search (30 trials)			
vs. Random Search (30 trials)	-0.38	0.705	-0.10
TPE Search (292 trials)			
vs. Random Search (292 trials)	16.03	< 0.001	1.33

Table 4: Two-sample t-test results and effect sizes of the Hyperparameters Optimization of ES-HyperNEAT on the MNIST classification task comparing the following measures: TPE Search (30, 292, 2013 trials), Random Search (30 and 292 trials).

Verbancsics and Harguess (2015)[4] achieved an accuracy of 23.90% using HyperNEAT based on LeNet-5 topology (1998) with a population size of 256 and 2500 generations.

Part 1 – Experimental Results

TPE Optimization Results on MNIST

- Validation dataset of 271 configurations
 - 50 best configs and their neighbors within a window of 2
 - 50 randomly sampled configurations, including their 4 neighbors
 - 200 randomly selected configs with the lowest objective values, including their 4 neighbors
- Allocated timeframe of 15 days
 - 30 replications per configuration
 - 199 configurations validated
- 72 out of validated configs surpassed Random Search (>22%)
- Persuasive TPE superiority and robustness

Part 1 – Experimental Results

TPE Optimization Results on MNIST

- Best configuration from TPE Search reaches 29% accuracy
 - Surpassing previous study [4] achieving 23.90% accuracy using smaller pop size (100 vs. 256) and fewer generations (20 vs. 2500)
- TPE vs. Random Search
 - Highly significant differences and large effect sizes
 - Outperforms on Mean, Median, and Best accuracies

Method	Mean (%)	Median (%)	SD (%)	Best (%)	Worst (%)
Hyperparameters Optimization for the MNIST Classification					
Random Search	11.18	10.00	2.62	22.00	10.00
TPE Search	17.41	19.00	4.26	27.50	10.00
TPE Search Best	23.40	23.25	2.07	28.00	20.00
TPE Validation	16.68	18.50	5.01	29.00	10.00
TPE Validation Best	20.95	20.75	2.41	29.00	17.00

Table 5: Accuracy metrics for the Hyperparameters Optimization of ES-HyperNEAT on the MNIST classification task, using Random Search (292 trials), TPE Search (2013 trials), TPE Search Best (30 replications of the best configuration found during the TPE search), TPE Validation (validation over 72 configurations), and TPE Validation Best (30 replications of the best configuration found during the validation of the TPE search).

Comparison	t-statistic	p-value	Cohen's d
Hyperparameters Optimization for the MNIST Classification			
TPE Search vs. Random Search	34.34	< 0.001	1.53
TPE Search Best vs. Random Search	20.28	< 0.001	3.42
TPE Validation Best vs. Random Search	20.93	< 0.001	3.72
TPE Validation Best vs. TPE Search	7.86	< 0.001	0.84
TPE Validation Best vs. TPE Search Best	-4.28	< 0.001	-1.11

Table 6: Two-sample t-test results and effect sizes of the Hyperparameters Optimization of ES-HyperNEAT on the MNIST classification task comparing the following measures: TPE Search (2013 trials), Random Search (292 trials), TPE Search Best (30 replications of the best configuration found during the TPE search), and TPE Validation Best (30 replications of the best configuration found during the validation of the TPE search).

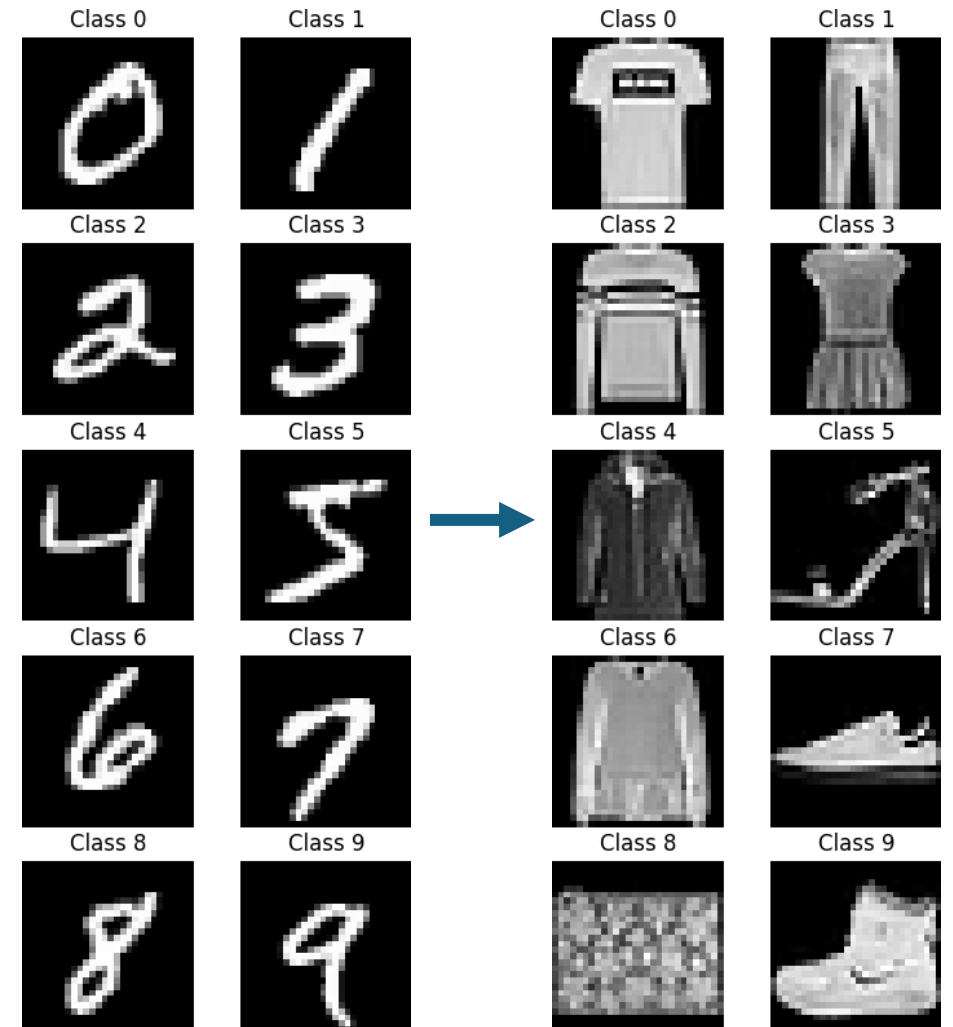
Part 2 – Experimental Setup: Transferability

- From: Best Configuration of
 - TPE Search & TPE Search Validation
- To: Varying Complexity tasks
 - Logic Operations & Fashion-MNIST
- Compare to Random Search

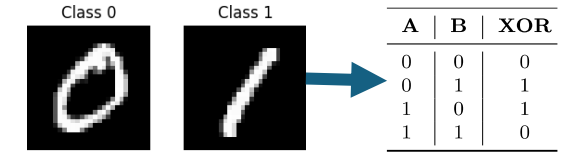
A	B	XOR	OR	AND	NAND	NOR	XNOR
0	0	0	0	0	1	1	1
0	1	1	1	0	1	0	0
1	0	1	1	0	1	0	0
1	1	0	1	1	0	0	1

Table 7: Truth table for six logic operations: XOR, OR, AND, NAND, NOR, and XNOR.

Random Search has the same setup for Fashion-MNIST [5] and Logic Operations. Only the number of generations for Logic Operations is reduced to 10.



Part 2 – Experimental Results: Transferability to Logic Operations



Method	Mean (%)	Median (%)	SD (%)	Best (%)	Worst (%)
XOR Operation					
Random Search	58.43	50.00	14.18	100.00	50.00
Best MNIST Config from TPE Search	83.33	83.25	1.38	87.47	79.13
Best MNIST Config from TPE Validation	78.34	77.61	3.33	83.29	75.00
OR Operation					
Random Search	60.11	55.21	35.73	100.00	25.00
Best MNIST Config from TPE Search	100.00	100.00	0.00	100.00	-
Best MNIST Config from TPE Validation	99.92	100.00	0.33	100.00	98.26
AND Operation					
Random Search	79.21	75.00	8.11	100.00	75.00
Best MNIST Config from TPE Search	87.91	87.50	1.11	91.15	87.41
Best MNIST Config from TPE Validation	86.54	87.05	1.12	87.50	83.87
NOR Operation					
Random Search	77.13	75.00	6.50	100.00	75.00
Best MNIST Config from TPE Search	86.65	81.17	8.56	100.00	76.01
Best MNIST Config from TPE Validation	84.82	81.25	8.62	100.00	75.00
XNOR Operation					
Random Search	57.67	50.00	13.19	100.00	50.00
Best MNIST Config from TPE Search	76.43	75.00	3.07	87.53	73.96
Best MNIST Config from TPE Validation	74.32	74.75	1.97	80.33	69.35
NAND Operation					
Random Search	56.35	75.00	30.40	100.00	20.00
Best MNIST Config from TPE Search	87.68	86.49	6.78	100.00	81.14
Best MNIST Config from TPE Validation	86.99	85.20	6.08	99.47	80.59

Table 8: Accuracy metrics for Logic Operations tasks, comparing Random Search (30 trials) and using the hyperparameters from the best MNIST configuration from the TPE search (30 replications) and the best MNIST configuration from the validation of the TPE search (30 replications).

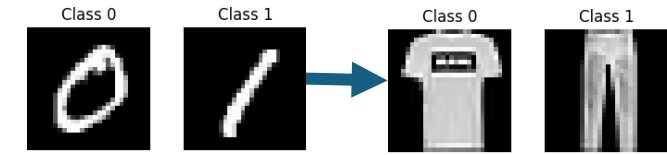
Comparison	t-statistic	p-value	Cohen's d
XOR			
Best TPE MNIST Config vs. Random Search	2.31	0.025	0.60
Best TPE Validation Config vs. Random Search	7.49	< 0.001	1.93
OR			
Best TPE MNIST Config vs. Random Search	4.14	< 0.001	1.07
Best TPE Validation Config vs. Random Search	6.10	< 0.001	1.58
AND			
Best TPE MNIST Config vs. Random Search	-0.43	0.663	-0.11
Best TPE Validation Config vs. Random Search	4.90	< 0.001	1.27
NOR			
Best TPE MNIST Config vs. Random Search	0.27	0.785	0.07
Best TPE Validation Config vs. Random Search	3.91	< 0.001	1.01
XNOR			
Best TPE MNIST Config vs. Random Search	0.71	0.483	0.18
Best TPE Validation Config vs. Random Search	6.84	< 0.001	1.76
NAND			
Best TPE MNIST Config vs. Random Search	3.66	0.001	0.94
Best TPE Validation Config vs. Random Search	5.41	< 0.001	1.40

Table 9: Two-sample t-test results and effect sizes for comparisons between the hyperparameters configuration from the best MNIST configuration from the TPE search (30 replications) and the best MNIST configuration from the validation of the TPE search (30 replications) with Random Search (30 trials) for each Logic Operation.

The best MNIST configs significantly outperform Random Search on XOR, OR, and NAND.

Part 2 – Experimental Results:

Transferability to **Fashion-MNIST**



- Best MNIST configs transfer
 - Significantly better than Random Search with large effect sizes
 - Higher Mean and Worst accuracies
- Demonstrates evidence of transfer within a similar domain
 - Even with higher intra-class variability and inter-class similarity

Method	Mean (%)	Median (%)	SD (%)	Best (%)	Worst (%)
Hyperparameters Optimization for the Fashion-MNIST Classification					
Random Search	11.60	10.00	3.08	23.50	10.00
Best MNIST Config from TPE Search	20.00	20.00	1.00	22.50	18.50
Best MNIST Config from TPE Validation	20.77	20.25	1.34	24.00	19.00

Table 10: Accuracy metrics for Fashion-MNIST classification task, comparing Random Search (292 trials) and using the hyperparameters from the best MNIST configuration from the TPE search (30 replications) and the best MNIST configuration from the validation of the TPE search (30 replications).

Comparison	t-statistic	p-value	Cohen's d
Hyperparameters Optimization for the Fashion-MNIST Classification			
Random Search (292 trials) vs. Random Search (30 trials)	0.47	0.641	0.10
Best MNIST Config from TPE Search vs. Random Search (30 trials)	11.42	< 0.001	2.95
Best MNIST Config from TPE Search vs. Random Search (292 trials)	32.75	< 0.001	2.85
Best MNIST Config from TPE Validation vs. Random Search (30 trials)	12.19	< 0.001	3.15
Best MNIST Config from TPE Validation vs. Random Search (292 trials)	30.13	< 0.001	3.09

Table 11: Two-sample t-test results and effect sizes for comparisons between the hyperparameters configuration from the best MNIST configuration from the TPE search (30 replications) and the best MNIST configuration from the validation of the TPE search (30 replications) with Random Search (30 and 292 trials) for the Fashion-MNIST classification task.

Conclusion

- TPE outperforms Random Search on MNIST classification task
- The best config identified by TPE achieves 29% on MNIST
 - 100% accuracy on Logics Operations with Random Search
- Transferability experiments suggest that effectiveness may depend on target task characteristics and similarity to source
 - MNIST to Fashion-MNIST appears to work
 - MNIST to Logics Operations less conclusive
 - Potential for hyperparameter bootstrapping

Future Work

- Alternative Optimization algorithms and combinations
 - (CMA-ES, NEAT, etc.)
- Refine and expand Search Space
- Additional tasks (simulations, statistical models, etc.)
- Deeper Transferability Investigation
 - Identify and adapt relevant source and target tasks
 - Advance Transfer Learning (meta-learning, etc.)
- Accelerate graph computations
- Goal is to develop auto-tuned evolving neural networks

Thank you for your attention

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References

- [1] Sebastian Risi and Kenneth O Stanley. 2010. Indirectly encoding neural plasticity as a pattern of local rules. In International conference on simulation of adaptive behavior. Springer, 533–543.
- [2] James Bergstra, Rémi Bardenet, Yoshua Bengio, and Balázs Kégl. 2011. Algorithms for Hyper-Parameter Optimization. In Advances in Neural Information Processing Systems, J. Shawe-Taylor, R. Zemel, P. Bartlett, F. Pereira, and K.Q. Weinberger (Eds.), Vol. 24. Curran Associates, Inc. https://proceedings.neurips.cc/paper_files/paper/2011/file/86e8f7ab32cfd12577bc2619bc635690-Paper.pdf
- [3] Peter I Frazier. 2018. A tutorial on Bayesian optimization. arXiv preprint arXiv:1807.02811 (2018).
- [4] Phillip Verbancsics and Josh Harguess. 2015. Image classification using generative neuro evolution for deep learning. In 2015 IEEE winter conference on applications of computer vision. IEEE, 488–493.
- [5] Han Xiao, Kashif Rasul, and Roland Vollgraf. 2017. Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms. arXiv preprint arXiv:1708.07747 (2017).